Musanze Harvest Intelligence Service

SWE 3513 — Individual Assignment 2 | Technical Report

Identity field	Value
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Verification code	AI-A2-8013

Decision problem

A potato cooperative field officer needs a reproducible tool to estimate harvest quantity, flag potentially high-risk fields, and group farms with similar measured conditions. This prototype generates synthetic records, trains three machine-learning tasks, and exposes predictions through a validated API.

Prediction questions

1. What harvest yield (tons) does the regression model estimate? 2. Does the classifier label the field as high risk? 3. What is the estimated high-risk probability? 4. Which unsupervised cluster is most similar to this record?

Ethical boundary

This is an educational decision-support prototype using synthetic data. It must not replace an agronomist or determine access to finance, insurance, food assistance, or public services. Human review is required for any consequential use.

2. Data and reproducibility

The personal seed is the final four registration digits (8013). NumPy default_rng(8013) generates 240 + (8013 mod 41) = 258 base records. Three exact duplicates are appended, resulting in 261 raw rows. The cleaner removes the three duplicate rows and retains 258 records. Missing values are deliberately inserted into soil_moisture_pct and fertilizer_kg, then imputed only within the training pipelines.

Column	Meaning / generation rule
rainfall_mm	Uniform 600–1400 mm
avg_temp_c	Uniform 12–26 °C
soil_moisture_pct	Uniform 20–80%; 5% missingness introduced
fertilizer_kg	Uniform 0–250 kg; 5% missingness introduced
field_size_ha	Uniform 0.2–4.0 ha
season	Random category A or B
cooperative_support	Random binary value 0 or 1
yield_tons	Specified formula + N(0, 0.80), clipped at 0.50
high_risk	1 if yield per hectare is below the median, else 0

Raw rows: 261. Clean rows: 258. Verification code: AI-A2-8013.

Dataset fingerprint

Verification code: AI-A2-8013
File: data/AI_A2_2528013.csv
SHA256: 9bceb231945195b6699d83051760dd2c704d695d7a6eb31c2885de1eba3fb4cf

Validation report

{"verification_code": "AI-A2-8013", "file": "data/AI_A2_2528013.csv", "rows": 261, "columns": 9, "duplicate_rows": 3, "missing_values": {"rainfall_mm": 0, "avg_temp_c": 0, "soil_moisture_pct": 13, "fertilizer_kg": 14, "field_size_ha": 0, "season": 0, "cooperative_support": 0, "yield_tons": 0, "high_risk": 0}, "range_checks": "passed for all observed non-missing values", "target_leakage_check": "passed; target columns are not in feature list", "missing_values_policy": "preserved for train-only pipeline imputation"}

3. Pipeline design and leakage control

run_all.py checks that the supplied seed matches the last four registration digits, then regenerates data, removes duplicates, validates the dataset, trains the supervised models, trains clustering, and rebuilds the required artifacts. The fixed split uses test_size=0.20 and random_state=8013.

Supervised preprocessing

The regression and classification models use scikit-learn Pipelines and ColumnTransformer. Numeric missing values are median-imputed and numeric inputs scaled with StandardScaler. Season is imputed using the most frequent category and encoded using OneHotEncoder(handle_unknown='ignore'). These transformations are fitted on the training subset only and saved with each final model.

Target leakage

The input features are rainfall, temperature, soil moisture, fertilizer, field size, season, and cooperative support. yield_tons and high_risk are targets, never classifier inputs. Yield per hectare is used only to create the high-risk label and is not a model feature.

Candidates

Regression: mean DummyRegressor baseline, LinearRegression, RandomForestRegressor. Classification: most-frequent DummyClassifier baseline, LogisticRegression, RandomForestClassifier. Clustering: median imputation, StandardScaler, and KMeans for K=2 through K=6 evaluated using silhouette score.

Determinism

The generator and models use the personal seed 8013. Re-run the full command and compare artifacts/data_fingerprint.txt to confirm reproducibility in the submission environment.

4. Regression and classification

Regression metrics on the held-out test set

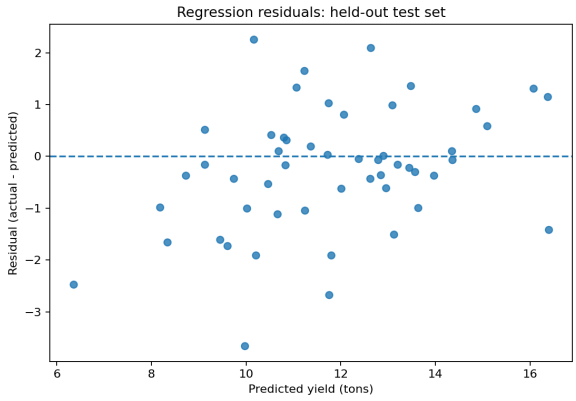
Model	MAE	RMSE	R^2
baseline	2.208	2.787	-0.002
linear_regression	0.924	1.218	0.808
random_forest	1.264	1.654	0.647

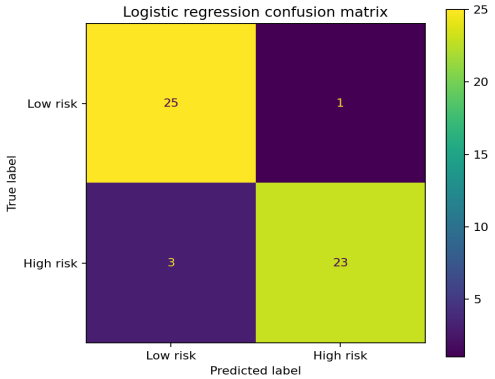
Selected regression model: linear_regression. Linear regression has the lowest held-out MAE and RMSE and the highest R2 among the evaluated regression candidates.

Classification metrics

Model	Accuracy	Precision	Recall	F1	ROC-AUC
baseline	0.500	0.000	0.000	0.000	0.500
logistic_regression	0.923	0.958	0.885	0.920	0.991
random_forest	0.923	0.958	0.885	0.920	0.964

Selected classifier: logistic_regression. Logistic regression matches random forest on accuracy, precision, recall, and F1 in this split, and has the higher ROC-AUC.





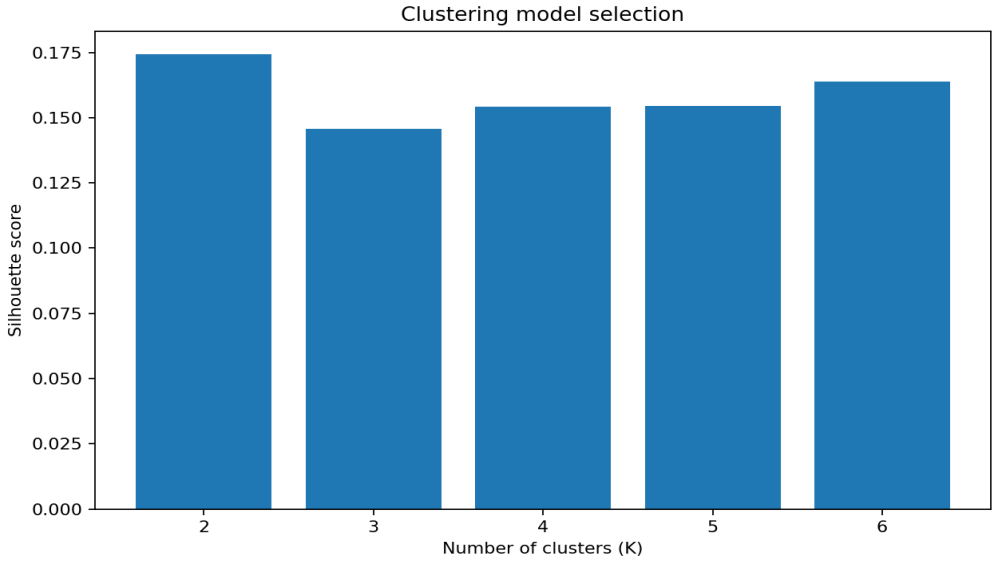
Student-authored error interpretation required: choose one row in artifacts/test_predictions.csv and explain its actual yield, prediction, absolute error, and a plausible reason for the difference. Identify one false positive or false negative from the confusion matrix and explain its practical significance.

5. Clustering and operational meaning

KMeans was evaluated for K=2…6. The selected K is 2; silhouette scores: {"2": 0.17443636260471487, "3": 0.14584768080161128, "4": 0.1540839798358165, "5": 0.1545560772405916, "6": 0.16376127855797523}.

The silhouette score is modest, so groups overlap and should not be interpreted as sharply separated types. Cluster profiles are averages of synthetic input features, not a farm-quality or risk score.

Cluster	avg_temp_c	cooperative_support	fertilizer_kg	field_size_ha	rainfall_mm	soil_moisture_pct
0	18.98	1.00	123.39	2.11	974.72	48.88
1	19.37	0.00	116.76	2.00	1013.63	51.41



Student-authored cluster interpretation required: choose one cluster, describe at least two feature means, compare it with another cluster, and explain why these synthetic profiles must not be used to rank real farms.

6. API and testing

Method / endpoint	Purpose
GET /health	Returns service health and verification code
GET /model-info	Returns task, model, and synthetic-data information
POST /predict	Accepts validated farm features and returns yield, risk, probability, cluster, verification code

Input contract

rainfall_mm: 600–1400; avg_temp_c: 12–26; soil_moisture_pct: 20–80; fertilizer_kg: 0–250; field_size_ha: 0.2–4.0; season: A or B; cooperative_support: 0 or 1. Missing fields, invalid types, out-of-range values, and extra fields are rejected by the API with HTTP 422.

Valid input example

{"rainfall_mm":1000,"avg_temp_c":20,"soil_moisture_pct":50,"fertilizer_kg":100,"field_size_ha":2,"season":"A","cooperative_support":1}

Invalid input example

{"rainfall_mm":2000,"avg_temp_c":20,"soil_moisture_pct":50,"fertilizer_kg":100,"field_size_ha":2,"season":"A","cooperative_support":1}

Automated tests

Nine tests cover health, model information, valid prediction, missing field, wrong type, range violation, repeatability, extra-field rejection, and synthetic-data disclosure. The prepared environment run completed with 9 passed. Re-run pytest -q on the machine used for submission and retain the actual output.

Commands: python run_all.py --student-id 25/28013 --seed 8013 ; python predict.py --json samples/valid.json ; python predict.py --json samples/invalid.json ; pytest -q ; uvicorn app:app --host 127.0.0.1 --port 8000

7. Model card and reflection

Intended use

Classroom demonstration of reproducible data generation, supervised learning, clustering, and validated API inference.

Out-of-scope use

Do not use these predictions to make or automate lending, insurance, public-service eligibility, or high-stakes farm allocation decisions. Do not treat clusters as quality labels or the predicted probability as certainty.

Data and performance limitations

All rows are synthetic and generated from a fixed mathematical formula. Artificial missingness does not represent real data collection. The held-out scores measure performance on this synthetic sample only and cannot demonstrate real-world agricultural accuracy. Real deployment would require representative field data, agronomist review, external validation, monitoring, and documented governance.

Student reflection — complete in your own words before submission

1. What did you learn from comparing the baseline with the selected models?

2. Which model error concerned you most, and why?

3. What would you change or test next, and why?

4. Which code component can you explain and modify independently?

Required pre-submission completion

Replace identity placeholders, write your own error and cluster interpretations, review the AI-use disclosure, add a real repository/commit hash if required, and record the required continuous unedited demo video in your actual environment.